

TASK-7 : GET BASIC SALES SUMMARY FROM A TINY SQLITE DATABASE USING PYTHON

I have created sales_data.csv Dataset.

1) Import The Necessary Library :

```
In [8]: import sqlite3
import pandas as pd
import matplotlib.pyplot as plt
```

2) Connect to the SQLite database :

```
In [9]: conn = sqlite3.connect('sales_data.db')
print("SQLite database connected successfully.")
```

SQLite database connected successfully.

3) Define and execute the SQL query :

```
In [10]: # Craft a SQL query to calculate the total quantity sold and total revenue per p

sales_summary_query = """
    SELECT
        product,
        SUM(quantity) AS total_quantity_sold,
        SUM(quantity * price) AS total_revenue
    FROM
        sales_data
    GROUP BY
        product
    ORDER BY
        total_revenue DESC;
    """
```

4) Load query results into a Pandas DataFrame :

```
In [11]: sales_summary_df = pd.read_sql_query(sales_summary_query, conn)
```

5) Display the sales summary :

```
In [12]: # Print the DataFrame to view the results.

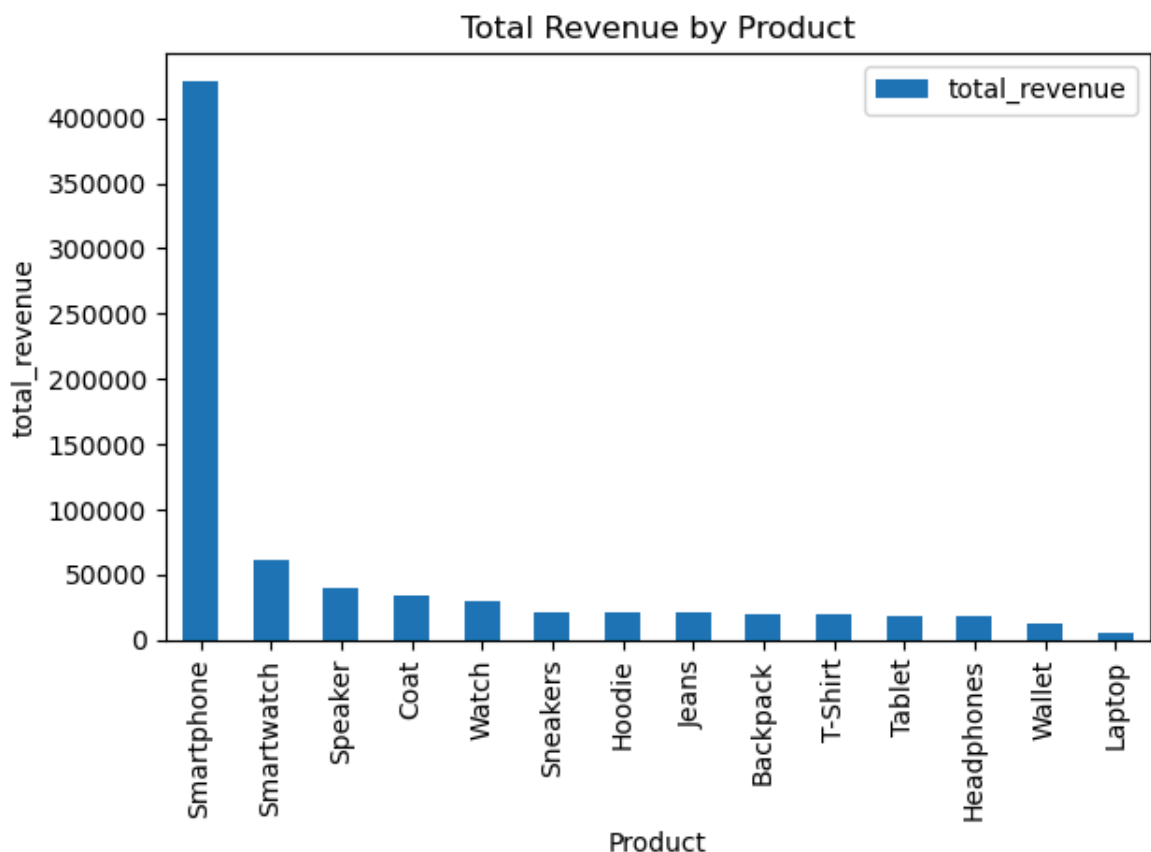
print("Basic Sales Summary:")
print(sales_summary_df)
```

Basic Sales Summary:

	product	total_quantity_sold	total_revenue
0	Smartphone	713	427800
1	Smartwatch	304	60800
2	Speaker	496	39680
3	Coat	338	33800
4	Watch	218	30450
5	Sneakers	270	21600
6	Hoodie	525	21000
7	Jeans	413	20650
8	Backpack	408	20400
9	T-Shirt	1005	20100
10	Tablet	45	18000
11	Headphones	180	18000
12	Wallet	440	12150
13	Laptop	5	6000

6) Create a bar chart using Pandas' built-in plotting functionality :

```
In [15]: sales_summary_df.plot(kind='bar', x='product', y='total_revenue', title='Total R
plt.xlabel("Product")
plt.ylabel("total_revenue")
plt.tight_layout() # Adjusts plot to prevent labels from overlapping
plt.show()
```



The top-performing product is the Smartphone with the highest revenue.

7) Save the chart as an image file :

```
In [16]: plt.savefig("sales_chart.png")
         print("\nChart saved as sales_chart.png")
```

Chart saved as sales_chart.png
<Figure size 640x480 with 0 Axes>

8) Close the database connection

```
In [17]: conn.close()
         print("\nDatabase connection closed.")
```

Database connection closed.

```
In [ ]:
```